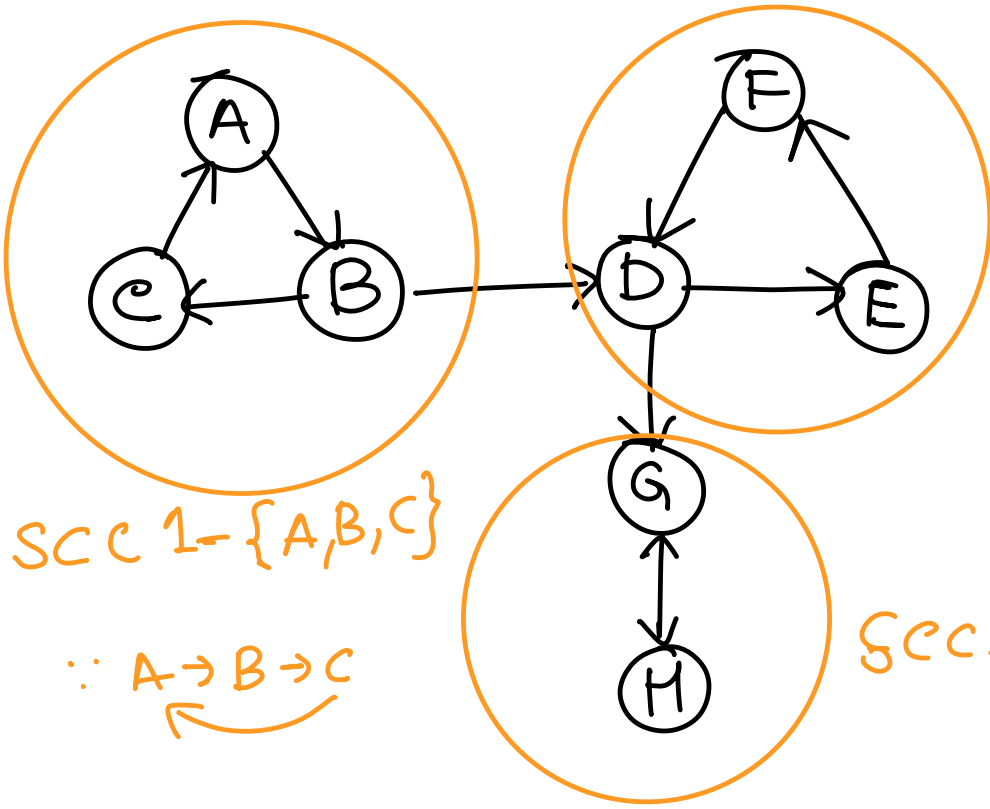
Strongly Connected Components (SCCs)



"Directed Graph"

SCC 1 = {A, B, C}

∵ A → B → C

SCC 2 = {D, E, F}

SCC 3 = {G, H}

Def^n: (Strongly Connected Component)

A SCC is a maximal vertex set C s.t every $x \in C$ can reach every $y \in C$ by a directed path.

visited [u] – True after u has recieved a discovery time

S – explicit active-vertex stack

OnStack [u] – Boolean value true exactly while u is currently in S.

Tarjan's Algorithm to find SCCs:

(I) Initialization

Tarjan - SCC(G)

1. $dt \leftarrow 0$ // discovery time
2. $S \leftarrow \phi$
3. For every vertex u:
   visited[u] ← False
   OnStack[u] ← False
4. For every vertex u:

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If visited[u] = false

Strong-Connect(u)

(II) Recursive Procedure:

Strong-Connect(u)

1. visited[u] ← True
2. dt ← dt + 1
3. d[u] ← dt
4. low[u] ← d[u]

5. Push u onto S

6. OnStack[u] ← True

7. For every outgoing arc
   $u \rightarrow v$ of Adj[u]

8. If visited[u] = False

9. : : Strong-connect(v)

10. : : low ← min{low[u], low[v]}

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11. else if OnStack[v] = true

12. ¦ ¦ low[u] ← min{low[u], d[v]}

13. else

14. ¦ do nothing . // v ∈ strongly connected component

15. If low[u] = d[u]

16. Start a new component

17. Repeat :

18. : $w \leftarrow Top(S)$

19. : Pop S

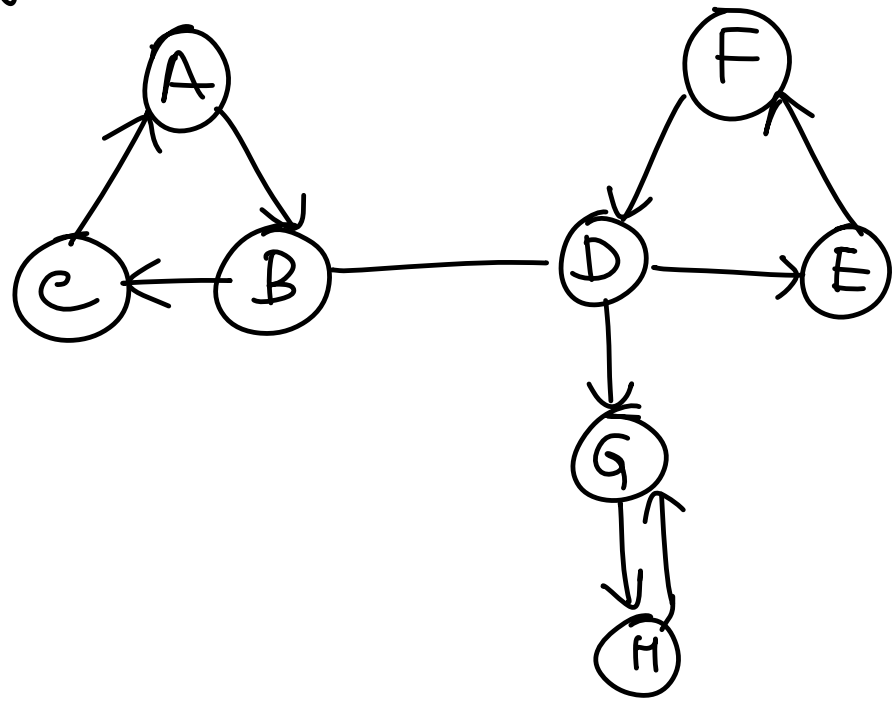
20. : onstack[w] $\leftarrow$ False

21. : Add w to C

22. until $w = u$

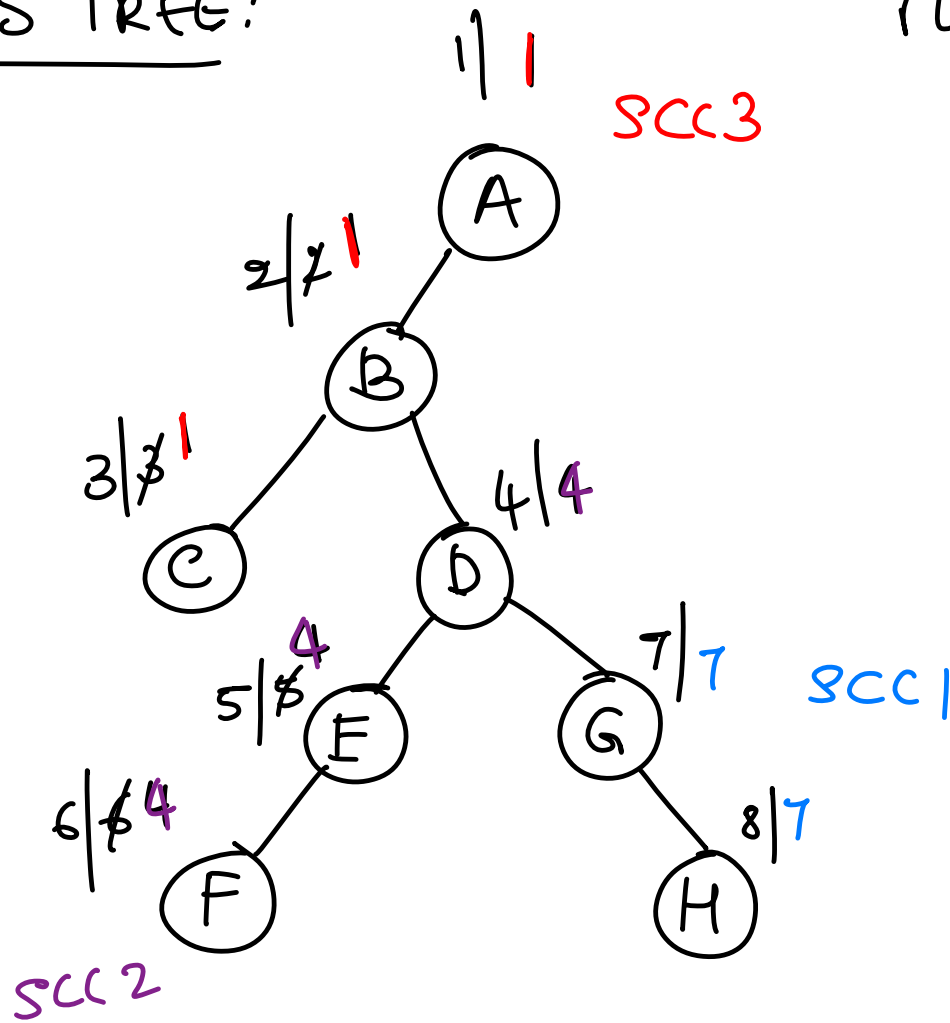
23. Output C

eg :



DFS TREE:

(lexicographic)



Step	Action	Active Stacks (Top is at right)	OnStack Status	Low link computation
1.	Discover A; Push A.	[A]	{A}: True	low[A] = d[A] = 1
2.	Discover B, Push B. Tree Arc A → B	[A, B]	{A, B}: True	low[B] = d[B] = 2
3.	Discover C Push C Tree Arc B → C	[A, B, C]	{A, B, C}: True	low[C] = d[C] = 3

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4.	Test A: Examine $C \to A$ (back arc edge)	$[A, B, C]$	$\{A,B,C\}$: True	OnStack[A] = true $low[C]$ $C \to A$, $u \to v$ $= \min\{low[C]^{u}, low[A]^{v}\}$ $= \min\{3,1\} = 1$
5.	Return to B	$[A, B, C]$	$\{A,B,C\}$: True	$B \to C$ $low[B] =$ $\min\{low[B], low[C]\}$ $= \min\{2,1\} = 1$
6.	Discover D Push D $(B \to D)$	$[A, B, C, D]$	$\{A,B,C,D\}$: True	$low[D] = d[D]$ $= 4.$
7.	Discover E push E tree arc $(D \to E)$	$[A,B,C,D,E]$	$\{A,B,C,D,E\}$: True	$low[E] = d[E]$ $= 5.$

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8.	Discover F Push F $(E \to F)$ tree arc	$[A, B, C, D, E, F]$	$\{A, B, C, D, E, F\}$: True	$low[F] = d[F] = 6.$
9.	Test D $(F \to D)$ back arc	$[A, B, C, D, E, F]$	D is active $\{A, B, C, D, E, F\}$: True	$F \to D$ $low[F] = \min\{low[F], d[D]\}$ $= \min\{6, 4\} = 4$
10.	Discover G Push G	$[A, B, C, D, E, F, G]$	$\{A, B, C, D, E, F, G\}$: True	$low[G] = d[G] = 7.$
11	Discover H Push H	$[A, B, C, D, E, F, G, H]$	$\{A, B, C, D, E, F, G, H\}$: True	$low[H] = d[H] = 8$

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12.	Test H (back arc H→G)	[A,B,C,D,E,F,G,H]	{A,B,C,D,E,F,G,H} : True G is active	H→G low[H] = min{low[H], low[G]} = min{8,7} = 7
13.	Return to G	[A,B,C,D,E,F,G,H}	{A,B,C,D,E,F, G,H}: True	low[G] = min{low[G], low[H]} = min{7,7} = 7 //
14.	Pop H	[A,B,C,D,E,F,G]	OnStack(H) = False (rest True)	———
15.	Pop G low[G] = low[H] Output = {G,H} SCC1	A,B,C,D,E,F]	OnStack(G) = False (rest True)	———

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16.	Return to F (E→F)	[A, B, C, D, E, F]	{A, B, C, D, E, F} True: A to F False: G, H	E→F low(E) = min{low[E], d(F)} = min{4, 6} = 4.
17.	Return to E E→D	[A, B, C, D, E, F]	{A B C D; E F} True: A to F False: G, H	low[D] = min{low(E), d(D)} = Min{4, 4} = 4.
18.	low[F] = low[E] = low[D]	[A, B, C]	True for only A, B, C	

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	Pop F, E, D. SCC2 = {D, E, F}			
19.	Return to B	[A, B, C]	True for only A, B, C	low[B] $^{B \to D}$ = min{low[B], d[D]} = min{1, 4} = 1.
20.	Return to A	[A, B, C]	True for only A, B, C	$A \to B$ low[A] = min{low[A], d[B]}

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				$\min\{1,2\} = 1$
21.	$low[c] = low[B] = low[A]$ Pop c	[A, B]	True for only A, B	———
22.	Pop B	A	True for A only	———
23.	Pop A	ϕ	False for all	———

Output SCC's = {c, B, A}

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